

ORAN ES SENIA, Algeria

WMO: 604900

Lat: 35.624N

Lon: 0.621W

Elev: 295

StdP: 14.54

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	36.8	39.1	30.4	24.8	51.2	33.6	28.5	49.7	30.4	58.9	27.6	59.3	1.7	180	0.443

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	18.4	93.9	70.1	91.0	70.6	88.0	71.0	77.2	85.6	75.9	83.9	74.8	82.4	12.0	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.8	131.8	81.9	73.5	125.8	80.9	72.0	119.7	79.8	40.7	86.0	39.5	83.9	38.5	82.4	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
25.5	21.6	18.8	DB	32.2	102.5	2.2	3.6	30.6	105.1	29.3	107.2	28.0	109.2	26.4	111.9
			WB	31.1	79.5	2.6	1.9	29.2	80.9	27.7	82.0	26.2	83.1	24.3	84.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	65.5	52.7	54.1	58.1	61.8	67.3	73.7	78.9	79.9	74.8	68.9	59.8	54.6
	DBStd	10.33	4.12	4.58	4.27	3.97	4.09	3.87	3.45	3.33	3.89	4.63	5.16	4.27
	HDD50	46	20	14	2	0	0	0	0	0	0	0	1	9
	HDD65	1549	382	305	218	111	23	0	0	0	0	17	170	323
	CDD50	5686	102	130	252	353	537	712	896	928	745	584	295	152
	CDD65	1715	0	1	3	14	94	262	431	463	296	136	14	1
	CDH74	14224	1	8	49	131	529	1917	4000	4564	2152	776	91	6
	CDH80	4925	0	3	8	18	101	551	1577	1855	624	172	15	1
(14) Wind	WSAvg	7.7	6.7	7.5	7.9	9.0	8.9	9.0	8.6	8.1	7.5	6.3	6.6	6.0
(15) Precipitation	PrecAvg	14.4	1.8	1.7	1.6	1.2	1.0	0.2	0.0	0.1	0.6	1.3	2.0	2.2
	PrecMax	24.8	5.7	5.5	5.0	4.5	5.8	1.4	1.1	0.8	2.8	6.7	7.2	7.5
	PrecMin	6.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	4.0	1.3	1.2	1.3	0.9	1.1	0.3	0.2	0.2	0.6	1.3	1.5	1.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.1	73.3	79.1	82.5	87.6	94.1	99.1	98.9	93.1	90.0	82.0	73.2
		MCWB	56.8	57.3	60.6	62.2	64.2	68.8	69.2	70.3	69.6	66.4	61.1	58.8
	2%	DB	66.9	69.4	73.7	77.4	82.6	89.3	93.4	94.7	88.2	84.3	76.2	68.4
		MCWB	56.0	56.5	58.8	60.5	63.7	68.3	70.0	71.1	69.9	66.7	61.9	57.7
	5%	DB	64.3	66.4	70.4	73.9	79.2	85.7	90.0	91.0	85.8	80.7	72.6	66.1
		MCWB	55.3	55.8	58.0	59.9	63.7	68.0	70.7	71.8	70.1	67.0	61.0	56.8
	10%	DB	61.7	63.9	67.8	71.3	76.7	82.6	87.5	88.0	83.8	78.4	69.6	63.5
		MCWB	54.3	55.2	57.1	59.0	63.2	68.0	71.5	72.6	69.9	66.5	60.1	56.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.5	60.5	63.8	65.3	69.3	74.7	78.3	79.5	76.0	72.3	67.2	61.7
		MCDB	65.7	67.6	74.3	76.0	80.1	84.3	86.9	87.5	84.2	80.3	73.9	69.2
	2%	WB	57.6	58.6	61.1	63.2	67.4	72.8	76.8	77.7	74.5	70.7	64.5	59.6
		MCDB	63.8	65.7	70.1	72.7	76.5	81.8	85.5	85.9	82.4	78.2	70.9	65.8
	5%	WB	56.4	57.2	59.4	61.8	66.1	71.7	75.5	76.5	73.3	69.2	62.9	58.1
		MCDB	62.4	63.9	67.3	70.4	74.6	80.0	83.5	84.5	80.9	76.8	69.7	64.1
	10%	WB	55.1	55.8	58.1	60.5	65.0	70.3	74.3	75.5	72.1	67.9	61.1	56.6
		MCDB	60.8	61.9	65.4	68.2	73.4	78.5	82.2	83.0	79.6	75.2	67.7	62.4
(35) Mean Daily Temperature Range	5% DB	MDBR	19.1	19.1	19.3	19.1	18.0	18.0	18.2	18.4	18.4	19.6	18.8	18.6
		MCDBR	22.4	23.3	24.7	24.6	22.7	22.7	23.3	23.4	21.2	23.1	22.9	21.6
		MCWBR	13.5	12.9	12.5	11.6	9.7	9.3	9.4	9.8	9.1	10.2	11.8	12.6
	5% WB	MCDBR	19.5	19.4	20.8	20.0	18.1	17.7	17.1	17.1	17.0	19.5	19.2	19.5
		MCWBR	12.7	12.0	11.6	10.5	8.9	8.4	8.3	8.4	8.6	10.0	10.9	12.4
(40) Clear-Sky Solar Irradiance	taub		0.327	0.353	0.413	0.429	0.462	0.488	0.536	0.510	0.468	0.416	0.368	0.335
	taud		2.490	2.413	2.217	2.198	2.129	2.060	1.933	2.008	2.137	2.277	2.400	2.484
	Ebn at Noon		277	280	270	272	264	256	242	246	250	254	257	266
	Edh at Noon		27	32	43	45	49	53	60	54	45	36	29	25
(44) All-Sky Solar Radiation	RadAvg		863	1130	1516	1836	2122	2358	2307	2077	1668	1280	900	776
	RadStd		72	109	100	134	144	83	59	66	62	84	74	54

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.38	N/A	N/A	N/A	+0.92	+0.86	N/A	N/A	+164	+115
Regional (10 neighbors)	+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page